UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which item is NOT a programming paradigm?
   A. Structured programming
   B. Procedural programming
   C. Object-oriented programming
   D. Debugging

2. How many times does a while loop execute when its condition is false initially?
   A. Zero
   B. Until the user stops it
   C. Two
   D. One

3. Which identifier is valid in C++?
   A. total score
   B. 2total
   C. int
   D. total_score

4. Which statement immediately exits a loop?
   A. break
   B. continue
   C. cin
   D. return 0

5. Which declaration creates a floating-point variable named price?
   A. bool price;
   B. float price;
   C. char price;
   D. int price;

6. Which operator reads data from cin?
   A. %
   B. =
   C. >>
   D. <<

7. What does the % operator return?
   A. The quotient
   B. The square root
   C. The absolute value
   D. The remainder of integer division

8. Which part of a for loop is executed after each iteration?
   A. The condition only
   B. The initialisation
   C. The update expression
   D. The loop body twice

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9. What is pseudocode?
   A. A structured English-like description of an algorithm
   B. A type of compiler error
   C. A C++ header file
   D. Machine code produced by an assembler

10. Which loop is commonly used to process every array element?
   A. break
   B. switch
   C. for
   D. if

PART B (60 MARKS)

QUESTION 1

This question tests the program development life cycle.

a) Define source code and object code.

(3 marks)

b) Describe THREE stages of the program development life cycle.

(2 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade: 80-100 A, 60-79 B, 50-59 C, 40-49 D, below 40 F.

a) Draw a flowchart for the grade classification.

(6 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(7 marks)

QUESTION 3

This question tests type conversion and constants.

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

(6 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

(5 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
   for (int j = 1; j <= i; j++) {
      cout << j;
   }
   cout << endl;
}
```

a) Trace the program and show the exact output.

(10 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(5 marks)

QUESTION 5

A shop stores the prices of five items in an array.

a) Declare an array named prices with the values 20.5, 55.0, 15.9, 75.0 and 10.0. Display all values.

(6 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(10 marks)

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PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

END OF QUESTION PAPER